

MSDS 535

Week 9 – Chapter 9

Overview and Objectives

- Understands cluster analysis advanced methods.
- Develop Probabilistic model-based clustering.
- Interpret Clustering high-dimensional data.
- Modify Biclustering.
- Select Dimensionality reduction for clustering.
- Select Clustering graph and network data.
- Compile Semisupervised clustering.
- Summarize cluster analysis advanced methods.

Probabilistic model-based clustering

- In many cluster analysis methods, data objects are typically assigned to a single cluster, which is necessary in applications like customer marketing.
- However, in some cases, rigid cluster assignments are not ideal.
- An example of this need is found in clustering product reviews for an e-commerce company.
- Ideally, reviews should belong to multiple clusters, reflecting their multifaceted topics.

Probabilistic model-based clustering

- Similarly, when studying user search intent in online stores, some sessions can belong to more than one cluster.
- This section explores clustering methods that allow objects to belong to multiple clusters.
- It begins with fuzzy clusters in Section 9.1.1 and extends to probabilistic model-based clusters in Section 9.1.2.
- The expectation-maximization algorithm, introduced in Section 9.1.3, provides a general framework for mining such clusters.

Fuzzy clusters

- A fuzzy set, denoted as S , is a subset of a set of objects X that allows each object to have a membership degree between 0 and 1.
- Formally, a fuzzy set can be represented as a function $FS : X \rightarrow [0, 1]$.
- Example 9.3 illustrates a fuzzy set in the context of digital camera popularity.
- It calculates the degree of popularity for a digital camera based on sales.
- Fuzzy clusters are introduced, where clusters are considered fuzzy sets of objects.

Fuzzy clusters

- A fuzzy clustering is represented using a partition matrix, ensuring membership degrees are between 0 and 1 for objects and clusters.
- An example (Example 9.4) demonstrates fuzzy clusters in an online store, grouping reviews into two fuzzy clusters.
- The evaluation of fuzzy clustering is discussed, measuring how well an object belongs to a cluster using distance and membership degrees.

Fuzzy clusters

- The sum of squared errors (SSE) is introduced as a measure of how well a fuzzy clustering fits a dataset.
- Fuzzy clustering allows objects to belong to more than one cluster, making it a soft clustering method.
- Traditional rigid clustering, where each object exclusively belongs to one cluster, is a special case of fuzzy clustering.

Probabilistic model-based clusters

- Fuzzy clusters, discussed in Section 9.1.1, offer flexibility by allowing objects to be part of multiple clusters.
- This section introduces a more general concept known as probabilistic model-based clusters to handle situations where objects can belong to multiple clusters in a probabilistic manner.
- Cluster analysis is based on the idea that the objects in a dataset belong to underlying categories, which are not directly visible and must be inferred from the data.

Probabilistic model-based clusters

- These hidden categories are referred to as probabilistic clusters, and they are represented by probability density functions describing how likely an instance of the cluster is found at a specific data point.
- Example 9.5 demonstrates the concept of probabilistic clusters using the example of digital camera categories and their associated probability density functions.
- A metric called the Sum of Squared Error (SSE) is introduced to evaluate how well a clustering fits the data.

Probabilistic model-based clusters

- Fuzzy clustering allows objects to be members of multiple clusters, offering more flexibility compared to traditional clustering, where each object belongs to just one cluster.
- Mixture models are introduced, where objects are considered as combinations of instances from multiple probabilistic clusters.
- The primary goal of probabilistic model-based cluster analysis is to discover a set of k probabilistic clusters that maximize the likelihood of generating the observed data.

Probabilistic model-based clusters

- This likelihood is usually represented as $P(\mathbf{D} \mid \mathbf{C})$, with $\mathbf{D}$ being the observed data and $\mathbf{C}$ being the set of probabilistic clusters.
- To make probabilistic model-based clusters more computationally manageable, parameterized probability density functions are often used.
- Example 9.6 provides an illustration of these concepts using univariate Gaussian mixture models with parameters like center (μ_j) and standard deviation (σ_j).

Expectation-maximization algorithm

- To compute fuzzy clusterings and probabilistic model-based clusterings, an approach based on the k-means algorithm is introduced.
- The k-means algorithm is a special case of fuzzy clustering and consists of two main steps in each iteration:
 - The Expectation Step (E-step): Objects are assigned to the nearest cluster center, expecting them to belong to that cluster.
 - The Maximization Step (M-step): Cluster centers are adjusted to minimize the sum of distances from the objects assigned to that cluster.

Expectation-maximization algorithm

- These steps can be generalized to tackle fuzzy clustering and probabilistic model-based clustering, forming an Expectation-Maximization (EM) algorithm:
 - The EM algorithm starts with an initial set of parameters and iterates until convergence.
 - Each EM iteration includes an E-step and an M-step.
- In the E-step, objects are assigned to clusters according to the current fuzzy clustering or probabilistic cluster parameters.
- In the M-step, new cluster parameters are found to minimize the Sum of Squared Error (SSE) in fuzzy clustering or the expected likelihood in probabilistic model-based clustering.

Expectation-maximization algorithm

- The EM algorithm aims to infer parameters that maximize the likelihood of generating the observed data.
- A practical example is given using the EM algorithm for fuzzy clustering and a univariate Gaussian mixture model for probabilistic model-based clustering.
- The EM algorithm may not always converge to the optimal solution and may require heuristics or multiple runs for better results, especially in complex scenarios.

Motivations of Clustering Analysis on High-Dimensional Data

- Some data objects have many attributes, creating high-dimensional data spaces.
- In high-dimensional data, traditional distance measures might not be effective in assessing object similarity.
- Challenges include irrelevant attributes, data sparsity, distance concentration effects, and optimization difficulties (the "curse of dimensionality").

High-Dimensional Clustering Models

- Clusters in high-dimensional data are often smaller and manifested by a smaller subset of attributes.
- One approach is to find clusters in subspaces, which can be axis-parallel or arbitrarily oriented.
- Subspace clustering methods aim to identify clusters in different subspaces.
- Categorization includes subspace clustering, projected clustering, and bi-clustering methods.
- Another approach is to use dimensionality reduction to transform objects to a lower-dimensional space for clustering analysis.

Subspace Clustering Methods

- Find clusters in subspaces of high-dimensional data.
- Data objects may belong to zero, one, or multiple clusters.
- Clusters may overlap in different subspaces.
- Examples include CLIQUE for grid-based methods.

Projected Clustering Methods

- Partition data into nonoverlapping subsets.
- Each data object belongs to exactly one cluster.
- Search for corresponding subspaces for each cluster.
- Example: PROCLUS.

Bi-Clustering Methods

- Cluster attributes and objects simultaneously.
- Used in biological data analysis and recommender systems.

Dimensionality Reduction Methods

- Construct new attributes to approach intrinsic dimensions.
- Transform objects to a lower-dimensional space for clustering analysis.

Axis-parallel subspace approaches

- CLIQUE
- PROCLUS
- Soft projected clustering methods
- Arbitrarily oriented subspace approaches

CLIQUE: A Subspace Clustering Method

- Subspace clustering methods discover clusters in various subspaces of high-dimensional data.
- CLIQUE is a representative method known for its grid-based approach.
- It employs a bottom-up strategy, initiating from low-dimensional subspaces and expanding to higher dimensions when necessary.
- Allows for overlaps among clusters in different subspaces.

PROCLUS: A Projected Clustering Method

- PROCLUS is a k-medoid-like method tailored for projected clustering in high-dimensional data.
- The method operates in three phases: initialization, iterative, and refinement.
- In the initialization phase, it generates potential cluster centers (medoids) through a greedy sample.
- The iterative phase improves medoid quality by evaluating attributes for each medoid and forming clusters.
- The refinement phase fine-tunes cluster subspaces using data objects assigned to a cluster.

Comparison of PROCLUS and CLIQUE

- PROCLUS assigns one object exclusively to a single cluster, making clusters non-overlapping.
- CLIQUE allows cluster overlaps, permitting one object to belong to multiple clusters.
- PROCLUS employs a top-down approach when searching subspaces, starting from the full space and ranking attributes. CLIQUE uses a bottom-up strategy.

Soft Projected Clustering Methods

- PROCLUS and similar methods use "hard" subspaces where an attribute is either included or excluded.
- Soft projected clustering methods, like LAC, introduce attribute weights to offer more flexibility and adapt to varying attribute importance and scaling.

Arbitrarily Oriented Subspace Approaches

- When clusters do not align well with attributes, arbitrarily oriented subspace approaches are employed.
- ORCLUS is an example of such a method, which partitions data into clusters and constructs subspaces based on the eigensystem of objects assigned to each cluster.
- ORCLUS uses weak eigenvectors to create new attributes and iteratively refines the eigensystem to achieve high-quality clustering.

Why and Where is Biclustering Useful?

- Biclustering techniques were initially developed to analyze gene expression data.
- Genes, residing on segments of DNA, play a vital role in the inheritance of traits in living organisms.
- Genes specify proteins and functional RNA chains, impacting an organism's cell development and genetic trait inheritance.

Why and Where is Biclustering Useful?

- Gene expression is the process by which genes produce functional products (RNA or proteins), and it's influenced by the genotype.
- Genotype represents the genetic makeup of cells, organisms, or individuals.
- Phenotype refers to observable characteristics of an organism.
- Gene expression data is represented in a matrix format, where rows correspond to genes, and columns represent samples or conditions.

Why and Where is Biclustering Useful?

- Biclustering allows for two-dimensional analysis of gene expression data:
 - Gene dimension: Treat genes as objects and samples/conditions as attributes.
 - Sample/condition dimension: Treat samples/conditions as objects and genes as attributes.
- Gene expression data analysis is essential in understanding mechanisms of diseases and clinical treatments.

Applications Beyond Gene Expression

- Biclustering isn't limited to gene expression; it has applications in other fields, such as recommender systems.
- For example, it can be used to analyze customer-product matrices in e-commerce for personalized recommendations.
- In these matrices, rows represent customers, columns represent products, and matrix elements reflect customer evaluations.
- Biclustering can identify customer groups with similar preferences or product groups with common customer interest.

Applications Beyond Gene Expression

- Biclusters usually exhibit characteristics such as:
 - Involvement of a small set of customers and products.
 - Some customers or products may be part of multiple clusters.
- Biclustering helps identify patterns and relationships in various data domains, making it a versatile tool.

Types of Biclusters

- Biclusters with constant values: Each element in the submatrix has a constant value.
- Biclusters with constant values on rows: Rows within the submatrix share a constant value, but values may differ between rows.
- Biclusters with constant values on columns: Columns within the submatrix share a constant value, but values may differ between columns.

Types of Biclusters

- Biclusters with coherent values: Elements in the submatrix exhibit synchronized changes across rows and columns, allowing for coherent patterns.
- Biclusters with coherent evolutions: Elements in the submatrix show coordinated changes over time without exact value constraints.

Biclustering Methods

- Biclustering methods aim to discover biclusters in noisy data. Two major types of methods exist:
 1. Optimization-based methods: Identify biclusters iteratively by searching for the submatrix with the highest significance score.
 2. Enumeration methods: Enumerate biclusters that meet specified noise tolerance criteria.

Biclustering Methods

- Enumeration methods like MaPle guarantee the completeness of results and do not miss overlapping biclusters.
- MaPle systematically enumerates combinations of conditions and finds maximal δ -pClusters, where δ controls the noise tolerance.
- Pruning techniques are employed in MaPle to speed up the search while retaining completeness.

Biclustering Methods

- These biclustering methods are critical for pattern discovery in various applications.
- Biclustering is a powerful approach for analyzing data patterns in gene expression and other applications, making it useful for understanding complex relationships within data sets.

Linear dimensionality reduction methods for clustering

- High-dimensional data clustering faces challenges due to data set characteristics.
- Challenges:
 - Correlation between attributes in high-dimensional data can lead to redundancy.
 - Different attribute observations may be recorded with varying scales and improper normalization.
- Dimensionality reduction aims to transform high-dimensional data into a lower-dimensional space while retaining important data properties.

Linear dimensionality reduction methods for clustering

- Example 9.12 illustrates clustering in a derived space for improved results in cluster separation.
- Dimensionality reduction, ideally with low dimensionality, can be extended to high-dimensional data.
- Principal Component Analysis (PCA) is a frequently used method for dimensionality reduction.
- PCA identifies meaningful basis vectors to re-express data and reduce correlation among attributes.

Linear dimensionality reduction methods for clustering

- PCA steps:
 1. Normalize the data.
 2. Compute eigenvectors of the covariance matrix.
 3. Choose top-k eigenvectors for dimensionality reduction based on cumulative eigenvalues.
- The covariance matrix reflects noise and redundancy in the original data representation.

Linear dimensionality reduction methods for clustering

- PCA aims to remove redundancy and ensure that dimensions in the new basis are uncorrelated.
- The top-k eigenvectors are used to reduce dimensionality while maximizing signal and minimizing noise.
- Example 9.13 demonstrates PCA applied to a 3-D data set, resulting in 2-D representation.
- PCA is a parameter-free method that can be applied to various numeric data sets.
- PCA's agnostic nature to data sources can be a limitation, as it does not consider domain-specific knowledge during dimensionality reduction.

Nonnegative matrix factorization (NMF)

- Nonnegative Matrix Factorization (NMF) clusters data by factorizing a matrix X into two nonnegative matrices, H and W , to represent the data as $X \approx HW$.
- H ($n \times k$) shows how n objects are assigned to k clusters, while W ($k \times m$) defines the cluster centers.
- NMF prevents negative entries in H to enhance interpretability.
- Applications include image clustering.

Nonnegative matrix factorization (NMF)

- NMF can relate to k-means clustering when H and W are orthogonal.
- Various cost functions like sum of squared errors and Kullback-Leibler divergence are used.
- NMF lacks an efficient exact solution but employs algorithms like the multiplicative update rule.

Nonnegative matrix factorization (NMF)

- The algorithm initializes W and H randomly and iteratively updates them.
- Challenges include sensitivity to initialization, selecting appropriate stopping criteria, and scalability concerns with large, high-dimensional datasets.
- NMF offers flexibility for addressing different clustering problems using varied objective functions and constraints.

Spectral clustering

- Spectral clustering is used for clustering and is based on the idea that in high-dimensional spaces, pairwise distances may not be reliable for clustering analysis.
- Spectral clustering relies on the close neighborhood relation between data points in a similarity graph.
- The process involves transforming the original data into a similarity graph, embedding the data in a new space, and extracting clusters using a classical algorithm like k-means.

Spectral clustering

- Spectral clustering constructs a similarity graph where vertices represent data points, and edges connect close neighbors.
- There are three common methods to form edges in the similarity graph: ϵ -neighborhood graph, nearest neighbor graphs, and fully connected graphs with Gaussian similarity functions.
- Spectral clustering uses Laplacian operators on the similarity graph, including unnormalized and normalized graph Laplacians, to embed data points in a new space.

Spectral clustering

- The first k eigenvectors of the Laplacian matrix are used to form the new space, where nodes collapse into clusters.
- Classical clustering algorithms, like k -means, are employed to extract clusters in the new space.
- Spectral clustering is particularly useful in high-dimensional applications like image processing but can be computationally expensive.
- It can be combined with other clustering methods for more robust results, making it a versatile tool in data analysis.

Clustering graph and network data

- Cluster analysis on graph and network data is crucial for extracting valuable knowledge and information from increasingly popular applications.
- A bipartite graph can represent customer purchase behavior, where one set of vertices represents customers, and the other represents products, with edges indicating purchases.
- Cluster analysis on bipartite graphs can lead to customer or product groupings for personalized recommendations.

Clustering graph and network data

- Web search engines use bipartite graphs to analyze user queries and web page click-through information, identifying similar queries and patterns.
- Web graphs help detect web communities, hubs, authoritative pages, and web spams.
- Social networks can also be represented as graphs, and customer clustering can be used for targeted promotions and relationship management.

Clustering graph and network data

- Similarity measures for graph clustering, such as SimRank, are based on geodesic distance, random walks, and structural context.
- Challenges specific to graph clustering include the development of new similarity measures and effective clustering models for data without explicit attributes.
- Geodesic distance measures the shortest path length between two vertices in a graph.

Clustering graph and network data

- SimRank is a structural context-based similarity measure that quantifies the similarity between vertices based on common neighbors.
- Similarity based on random walk considers the likelihood that random walkers starting from two vertices reach the same target, measuring closeness by the probability of simultaneous receipt of messages.
- Personalized PageRank models the probability that a random walk starting from one node ends at another.
- Topical PageRank extends PageRank to consider topics and topic-specific probabilities, measuring similarity between nodes with respect to various topics.

Graph clustering methods

- Clustering on a graph involves partitioning the graph into clusters.
- The goal is to find well-connected vertices within clusters and weaker connections between clusters.
- A graph $G = (V, E)$ is partitioned by a cut, $C = (S, T)$, where $V = S \cup T$ and $S \cap T = \emptyset$.
- The cut set consists of edges $\{(u, v) \in E \mid u \in S, v \in T\}$.
- The size of a cut is the number of edges in the cut set.

Graph clustering methods

- In weighted graphs, the cut size is the sum of edge weights.
- Minimum cuts are important for deriving clusters.
- A minimum cut is one where the cut size is not greater than any other cut.
- Polynomial time algorithms exist to compute minimum cuts.
- However, using a minimum cut may not lead to effective clustering.

Sparsest Cut and Sparsity

- The concept of sparsity in a cut, $C = (S, T)$, is defined as the cut size divided by the minimum of $|S|$ and $|T|$.
- A cut is "sparsest" if its sparsity is not greater than any other cut.
- Multiple sparsest cuts can exist.
- The objective of a sparsest cut is to minimize the number of edges crossing partitions and balance partition sizes.

Modularity in Clustering

- Clustering quality is assessed using modularity (Q).
- Modularity measures the difference between observed and expected edges within clusters.
- Modularity (Q) formula: $Q = \sum (l_i * d_i) / (2 * |E|)$, $i=1$ to k , where l_i is edges within the i th cluster and d_i is the sum of degrees in the i th cluster.
- Optimal clustering maximizes modularity.

Challenges in Graph Clustering

- Challenges include high computational costs, sophisticated graph structures, high dimensionality, and sparsity.
- Efficient, scalable, and quality trade-offs are essential in large graphs.

Generic High-Dimensional Clustering Methods

- Extract similarity matrices from graphs.
- Apply generic clustering methods for high-dimensional data.
- Spectral clustering is often used to approximate optimal graph cut solutions.

Specific Clustering Methods by Searching Graph Structures

- Examples include SCAN (Structural Clustering Algorithm for Networks).
- Identify well-connected components as clusters.
- Grow clusters from core vertices based on structural context similarity.
- Similar to DBSCAN for density-based clustering.
- Handle outliers and hubs.

Probabilistic Graphical Model-Based Methods

- Treat graphs as observations generated by probabilistic models.
- Use heuristics on community generation to identify clusters.
- Utilize directed, undirected, or hybrid graphical models.
- Stochastic Block Model (SBM) is an example.

Semisupervised clustering

- In conventional clustering, data objects are not labeled, but sometimes domain knowledge is available.
- Semisupervised clustering addresses scenarios with some domain knowledge.
- This knowledge enhances clustering results.

Clustering with Partially Labeled Data

- In scenarios with partially labeled data, a small portion is labeled into categories.
- Using the labeled data to build a classifier may not be effective due to its small proportion.
- Semisupervised clustering methods use this labeled data to strengthen clustering results.

Constrained k-Means for Labeled Data

- Extends k-means with labeled data.
- Compute cluster means for labeled data.
- Assign unlabeled objects to clusters based on mean proximity.
- Iteratively update cluster means.

Seeded k-Means for Error-Prone Labeled Data

- Similar to constrained k-means but doesn't strictly trust labels.
- Assigns objects to the cluster whose mean is closest.
- Useful when labeled data may contain errors.

Clustering on Pairwise Constraints

- Domain knowledge from pairwise relations between objects.
- Must-link constraints group objects together.
- Cannot-link constraints keep objects apart.
- COP-k-means revises k-means to respect constraints.
- Assigns objects to clusters that respect constraints.
- Does not require backtracking.

Handling Soft Constraints with PCKmeans

- Soft constraints impose penalties on constraint violations.
- PCKmeans combines clustering quality and penalty scores in the objective function.
- Minimizes sum of squared error and penalty scores for must-link and cannot-link constraints.

More Types of Background Knowledge

- Semisupervised Hierarchical Clustering:
 - Use constraints with context information.
 - Apply constraints to hierarchical clustering dendrograms.
- Clusters Associated with Outcome Variables:
 - Discover clusters associated with specific outcome variables.
 - Can guide the formation of clusters in the presence of output variables.

More Types of Background Knowledge

- Active and Interactive Learning for Semisupervised Clustering:
 - Active learning acquires must-link and cannot-link constraints through user interaction.
 - Users provide feedback to adjust clustering during the iterative process.

Summary

- Conventional cluster analysis assigns objects exclusively to one cluster.
- Fuzzy clustering and probabilistic model-based clustering allow objects to belong to one or more clusters, using partition matrices to record membership degrees.
- Probabilistic model-based clustering estimates cluster parameters by treating data as observed samples.
- Mixture models represent data as a mix of instances from probabilistic clusters, generated through an expectation-maximization algorithm.

Summary

- High-dimensional data presents challenges due to irrelevant attributes, data sparsity, and distance concentration.
- Clustering for high-dimensional data focuses on axis-parallel and arbitrarily-oriented subspaces, employing subspace clustering, projected clustering, biclustering, and dimensionality reduction methods.
- Biclustering clusters objects and attributes simultaneously, aiming for constant values or coherent evolutions.
- Dimensionality reduction transforms high-dimensional data into lower-dimensional spaces to retain meaningful properties.

Summary

- Principal component analysis (PCA) identifies meaningful data basis, Nonnegative Matrix Factorization (NMF) decomposes data into non-negative matrices, and spectral clustering constructs new dimensions using an affinity matrix.
- Clustering for graph and network data involves measuring similarity, considering geodesic distance, structural context, random walk, and SimRank.
- Graph clustering can be modeled as computing graph cuts, optimizing sparsest cuts, or using modularity to assess quality.
- Challenges in graph clustering include high computational cost, sophisticated graphs, high dimensionality, and data sparsity.

Summary

- Generic graph clustering methods extract similarity matrices and apply clustering techniques.
- SCAN is a graph clustering algorithm searching graph structures for well-connected components.
- Probabilistic graphical model-based methods use heuristics for community generation, employing the stochastic block model (SBM).
- Semisupervised clustering uses domain knowledge, such as partially labeled data and pairwise constraints, to enhance clustering results.